Rowen Seykora

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Summary

Life sciences professional with a foundation in experimental microbiology and quantitative approaches to human disease. Interested in how microbial and host factors shape infection across molecular, community, and population levels, and in research that bridges mechanistic inquiry with systems-level analysis of infectious disease.

Education

Rochester Institute of Technology (RIT) — Rochester, NY, USA
B.S. in Molecular Biosciences — *Magna Cum Laude*

Publications

[1] Pending citation update - A systems serology study investigating correlates of protection against infection by *Vibrio cholerae*. 2023.

[2] Pending citation update - The high-throughput identification of RNA sequence elements that drive localization of transcripts in neuronal cells. 2022.

[3] Charles, R.C., Kelly, M., Tam, J.M., Akter, A., and 27 others including **Seykora, R.**, Harris J.B., and Ryan E.T. Humans Surviving Cholera Develop Antibodies against *Vibrio cholerae* O-Specific Polysaccharide That Inhibit Pathogen Motility. *mBio*, 2020. doi:10.1128/mbio.02847-20

[4] Marquis, K.A., **Seykora, R.**, Weiss, A.N., Morris, M.C., and Ferran, M.C. The VSV matrix protein inhibits NF-kB and the interferon response independently in mouse L929 cells. *Virology*, 548, 117–123, 2020. doi:10.1016/j.virol.2020.06.013

[5] Weil, A.A., **Seykora, R.**, Harris, J.B. *Vibrio cholerae* at the Intersection of Immunity and the Microbiome. *mSphere*, 4, 2019. doi:10.1128/mSphere.00597-19

[6] Pending citation update - A study investigating the presence of *Vibrio cholerae*-killing antibodies in young children and the impact on protection from future infection. 2019.

Presentations

[1] Pending citation update - A presentation examining vibriocidal antibody titers in children, and whether these are associated with protection against cholera similarly across age strata. *Session: 133. Enteric Infections, IDWeek.* San Francisco, California, USA. 2018.

[2] **Seykora, R.**, Marquis K., Ferran M.C. Differential effects of the vesicular stomatitis virus matrix (M) protein on NF-kB activation in L929 and HeLa cells. *American Society for Virology (ASV) 36th Annual Meeting.* Madison, Wisconsin, USA. 2017.

Technical Communication Experience

Science Writer (Community Contributor) — 2026 – Present
The Life Raft Group (patient advocacy & research for rare cancers such as GIST and liposarcoma)
- Synthesize a growing body of research on gastrointestinal stromal tumors (GISTs), a class of soft tissue sarcomas, translating complex biomedical literature into accessible prose for newly diagnosed patients and their family members.
- Author patient-facing content that pairs technical rigor with empathy, promoting research participation and strengthening patients' understanding of their condition.
- Serve as a liaison among cancer biologists, clinicians, and patient networks, disseminating research findings from their source to those most directly affected while maintaining scientific accuracy and accessibility.

Technical Consultant in Life Sciences & Technology — 2023 – 2025
Thermo Fisher Scientific; Raynmaker (agentic AI for animal health)
- Advised researchers and small-business clients on life sciences and technology needs, translating complex technical offerings into clear, tailored solutions.
- Managed structured CRM records for thousands of contacts, sustaining long-term engagement with evolving research needs.

Science Communication Specialist — 2022
Slingshot Biosciences
- Led content development for the launch of flow cytometry and synthetic cell biology reagents, producing educational writing, scientific figures, application notes, and articles that conveyed the scientific basis and applications of these technologies to both research and applied audiences.

- Collaborated cross-functionally with the R&D team, applying data interpretation and scientific literature review to distill experimental findings into nuanced, plain-language explanations that directly informed and strengthened written communications.

Research Experience

Molecular Biology Research Associate 2020 – 2021

University of Colorado, Anschutz Medical Campus

- Contributed to research in neuronal cells validating and examining active oligonucleotide sequences within 3′ UTRs found to be sufficient and necessary to drive localization of their transcripts to neurites. This established the groundwork for identification of RNA-binding proteins (RBPs) such as Unk (Unkempt), as associated with relevant sequence elements and required for efficient localization of the studied transcripts to neurites.
- Built upon a workflow for smFISH (fluorescence in situ hybridization) image data capture and analysis, improving the accuracy and completeness of spatial visualization of relevant RNAs in cells with complex, elongated morphology.
- Communicated research findings through presentations and written reports for both scientific peers and a broader community of molecular biologists, translating technical results into clear, structured narratives.

Microbiology Research Technician II 2017 – 2019

Massachusetts General Hospital (MGH), Division of Infectious Diseases

- Created a comprehensive epidemiological database by consolidating and analyzing pertinent information for approximately 800 cholera patients and their 1700+ household contacts collected over the course of two decades.
- Contributed to an ongoing systems serology-based initiative to identify novel humoral correlates of protection against infection with *Vibrio cholerae* by providing expertise in high-throughput analysis of results and by performing a variety of assays, such as Ig Luminex, ELISA, and antibody-dependent cellular phagocytosis assays, that measure functional activity of antibodies in convalescent cholera-patient serum.
- Conducted serosurveillance of cholera-endemic areas, including several sites in Haiti and Dhaka, Bangladesh, through high-throughput sample evaluation by real-time qPCR, multiplex PCR, ELISA, and vibriocidal titer assays.
- Mentored and instructed research students, and guided incoming research technicians throughout their initial training. Devised new protocols, upgraded existing methods, and authored highly detailed SOPs. Managed working order of the lab by procuring necessary reagents and supplies, maintaining and purchasing new equipment, and cataloging samples received by the lab.

Molecular Virology Undergraduate Researcher 2015 – 2017

Thomas H. Gosnell School of Life Sciences, Eukaryotic Viral Genetics Lab

- Optimized, performed, and interpreted assays including qPCR, ELISA, and immunofluorescence confocal microscopy to gauge the effect of vesicular stomatitis virus (VSV) on crucial host functions such as mRNA transcription and bidirectional nucleocytoplasmic transport.
- Conducted *in vitro* infections using an array of VSV mutants in multiple cell types to determine the cell-specific effects of VSV on the host, the mechanisms by which VSV evades the innate interferon (IFN) response to viral infection, and the specific viral protein(s) implicated in these immune evasion mechanisms.
- Propagated and titered virus stocks, cultured all cell lines used for infections, prepared all in-house reagents and solutions, and monitored laboratory conditions to ensure consistent experimental performance.
- Co-authored abstracts, student-faculty grants, presentations, and posters, and presented research findings at symposia and conferences.
- Mentored incoming research students and trained them in essential laboratory skills.

Core Competencies

BSL-II+ Protocol, Microbial Lab Techniques (Culture, Staining, Plaque Assay), Confocal and Epifluorescence Microscopy, Genetic Engineering Techniques (PCR, DNA Sequencing), Gene Expression Assays (Real-Time qPCR, ELISA, Luminex), Macromolecule purification (RNA, DNA, mAB, General Protein), R Programming and Stata MP + SPSS, Flow Cytometry (Sample Prep, Run, and Analysis).